The operation of finding the direct connection between $x$ and $y$ without the presence of the auxiliary variable $t$ is called **elimination of the parameter**. The question as to why should we deal with parametric equations is that two or more variables are reduced to a single variable, $t$.

For example, the equation of a circle with centre (0, 0) and radius $r$ is $x^2 + y^2 = r^2$ This equation describes the relationship between $x$ and $y$ and the equations

$x = r\cos t$ ; $y = r\sin t$ are parametric equations.

Conversely, if we eliminate $t$, we get $x^2 + y^2 = r^2$

If $y$ is regarded as a function of $x$ then $\dfrac{dy}{dx} = \dfrac{\frac{dy}{dt}}{\frac{dx}{dt}} = \dfrac{g'(t)}{f'(t)}$.

If we regard $x$ as a function of $y$, then the derivative of $x$ with respect to $y$ is,

$$\frac{dx}{dy} = \frac{\frac{dx}{dt}}{\frac{dy}{dt}} = \frac{f'(t)}{g'(t)}.$$

In the case of the circle, then $\dfrac{dy}{dx}$ is the slope of the tangent to the circle namely

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{r\cos t}{-r\sin t} = -\cot t \ .$$

**Example 10.26**

Find $\dfrac{dy}{dx}$ if $x = at^2$ ; $y = 2at$, $t \neq 0$.

**Solution**

We have $x = at^2$ ; $y = 2at$

$$\frac{dy}{dx} = \frac{y'(t)}{x'(t)} = \frac{2a}{2at} = \frac{1}{t}.$$

**Example 10.27**

Find $\dfrac{dy}{dx}$ if $x = a(t - \sin t), y = a(1 - \cos t)$.

**Solution**

We have $x = a(t - \sin t), y = a(1 - \cos t)$.

Now $\dfrac{dx}{dt} = a(1 - \cos t)$ ; $\dfrac{dy}{dt} = a\sin t$

Therefore, $\dfrac{dy}{dx} = \dfrac{\frac{dy}{dt}}{\frac{dx}{dt}} = \dfrac{a\sin t}{a(1-\cos t)} = \dfrac{\sin t}{(1-\cos t)}$ .



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